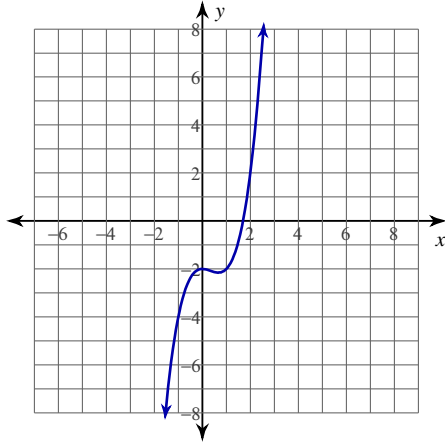


Normal Lines

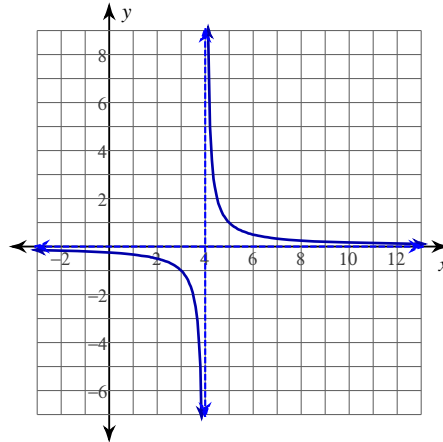
Date _____ Period _____

For each problem, find the equation of the line normal to the function at the given point. If the normal line is a vertical line, indicate so. Otherwise, your answer should be in slope-intercept form.

1) $y = x^3 - x^2 - 2$ at $(1, -2)$



2) $y = \frac{1}{x-4}$ at $(5, 1)$



3) $y = -x^3 + 15x^2 - 72x + 110$ at $(4, -2)$

4) $y = \frac{2}{x-3}$ at $(5, 1)$

5) $y = \frac{3}{x+2}$ at $\left(4, \frac{1}{2}\right)$

6) $y = (2x-8)^{\frac{1}{3}}$ at $(0, -2)$

7) $y = \ln(x+4)$ at $(-3, 0)$

8) $y = -\sin(2x)$ at $\left(-\frac{\pi}{2}, 0\right)$